Avalanche Analysis Tutorial

Zack Niedzwiecki

Spatial and Statistical Analysis of Avalanches Observed by the Gallatin National Forest Avalanche Center (GNFAC)

The GNFAC is the avalanche research and forecasting center for the Gallatin valley in Montana. In order to collect and distribute accurate data, they have split their forecast area into 8 different regions. These regions act independently as they have different characteristics. Each day, a forecast is shared with the public for each region and when avalanche data is collected it is paired with a region. In this tutorial, you will analyse the spatial distribution and behaviors of avalanches in each region of the GNFAC.

Github Repo: https://github.com/zackniedz/Stats_408_Final

Section 1: Preparation

Lets load packages, and clean the data to prepare it for analysis

```
# load libraries and data
# READ .kml
#install.packages("prettymapr")
#install.packages("gt")
library(sf)
library(tidyverse)
library(ggplot2)
library(ggspatial)
library(prettymapr)
library(gt)
library(patchwork)
library(dplyr)
#read in region shapefile recieved from GNFAC

reigons <- suppressMessages(
```

```
  suppressWarnings(
    sf::st_read("GNFAC Polygons ALL.kml", quiet = TRUE)
  )
)
```

Clean the data and take a look at it

```
#load GNFAC avalanche data

avy_data <- read.csv("GNFAC_2024-2025_AVY_DATA(Sheet1).csv")

#clean data to contain relevent collumns

avy1<-avy_data%>%
  select("Date", "Longitude", "Latitude", "Region","Elevation","Aspect", "Avalanche.Type","D
avy1 <- na.omit(avy1)

# manke it pretty

head(avy1) %>%
  gt() %>%
  tab_header(title = "Raw Avalanche Data (GNFAC)")
```

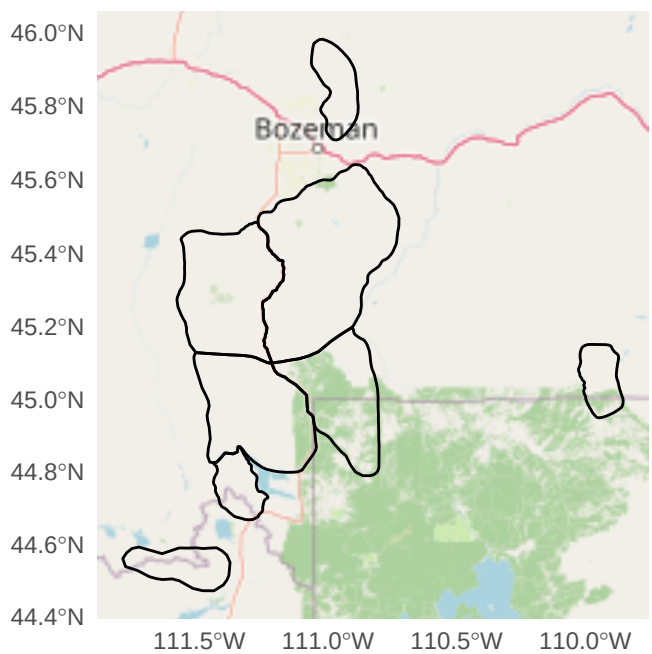
Raw Avalanche Data (GNFAC)

Date	Longitude	Latitude	Region	Elevation	Aspect	Avalanche.Type	D
5/3/2025	-110.923	45.8156	Bridger Range	7800	E	Wet loose-snow avalanche	
5/1/2025	-110.930	45.8084	Bridger Range	8000	E	Wet loose-snow avalanche	
4/19/2025	-110.976	45.9196	Bridger Range	7700	NE	Soft slab avalanche	
4/19/2025	-109.928	45.0722	Cooke City	10200	NE	Soft slab avalanche	
4/9/2025	-110.958	45.9043	Bridger Range	9000	SE	Soft slab avalanche	
4/7/2025	-109.938	45.0202	Cooke City	1000	SE	Wet loose-snow avalanche	

Section 2: Map the forecast zones and avalanche events

plot the forecast zones on a map

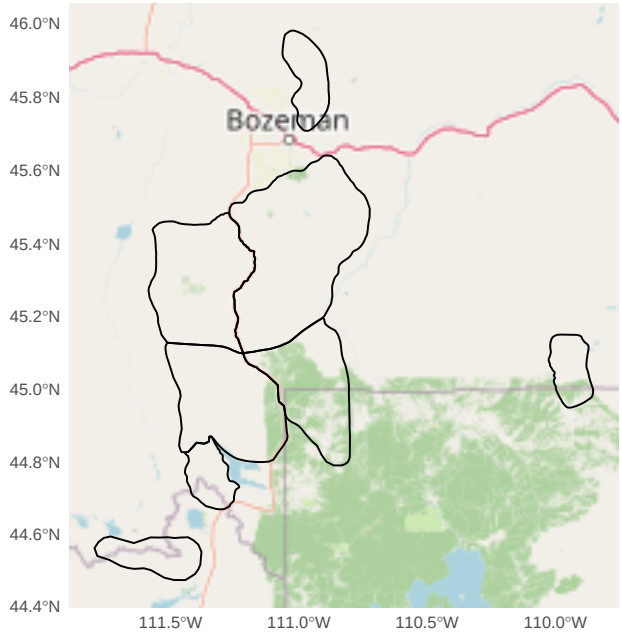
```
#Make map of shapefile and visualize

#set CRS
reigons <- st_transform(reigons, 4326)

#set plot conditions
pollygons<- ggplot() +
  annotation_map_tile(type = "osm") +   # basemap
  geom_sf(data = reigons, color = "black", linewidth = 0.5, alpha = 0) +
  coord_sf() +
  theme_minimal()
pollygons
```

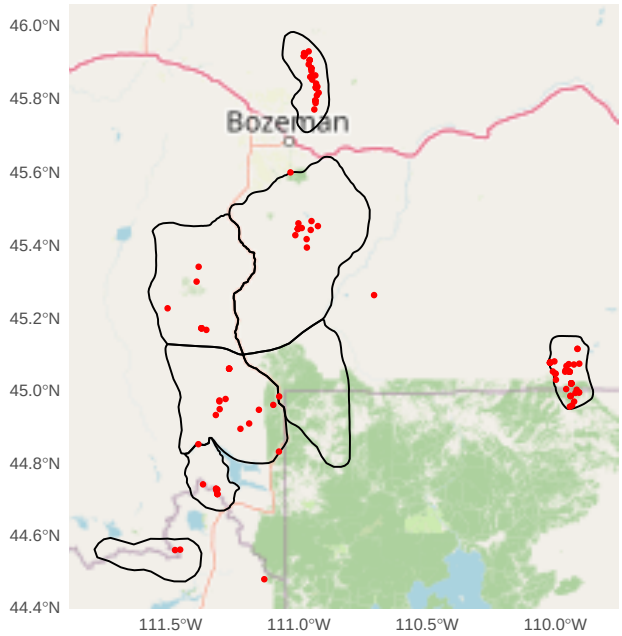


plot the observed avalanche events on the previously made map and stack them for side by side analysis.

```
#plot avalanches on map

#georeference the lat and long collumns of the .csv
```

```r
avy_map <- st_as_sf(avy1,
                    coords = c("Longitude", "Latitude"),
                    crs = 4326)  # WGS84

# plot the location of the avalanche events over the shapefile

point_map <- ggplot() +
  annotation_map_tile(type = "osm") +   # basemap
  geom_sf(data = reigons, color = "black", linewidth = 0.5, alpha = 0) +
  geom_sf(data = avy_map, color = "red", size = 1)+
  theme_minimal()

# stack the graphs

pollygons / point_map
```


46.0°N
45.8°N
45.6°N
45.4°N
45.2°N
45.0°N
44.8°N
44.6°N
44.4°N
Bozeman
111.5°W
111.0°W
110.5°W
110.0°W

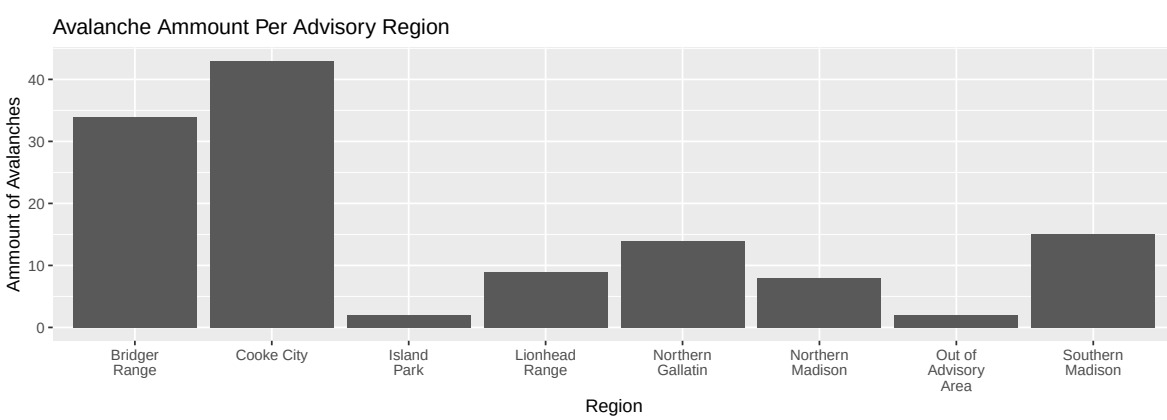

46.0°N
45.8°N
45.6°N
45.4°N
45.2°N
45.0°N
44.8°N
44.6°N
44.4°N
Bozeman
111.5°W
111.0°W
110.5°W
110.0°W

Section 3: Make plots of avalanche characteristics per region

The following chunks will make bar graphs of total events per zone

3A) make a plot of total avalanche events per region

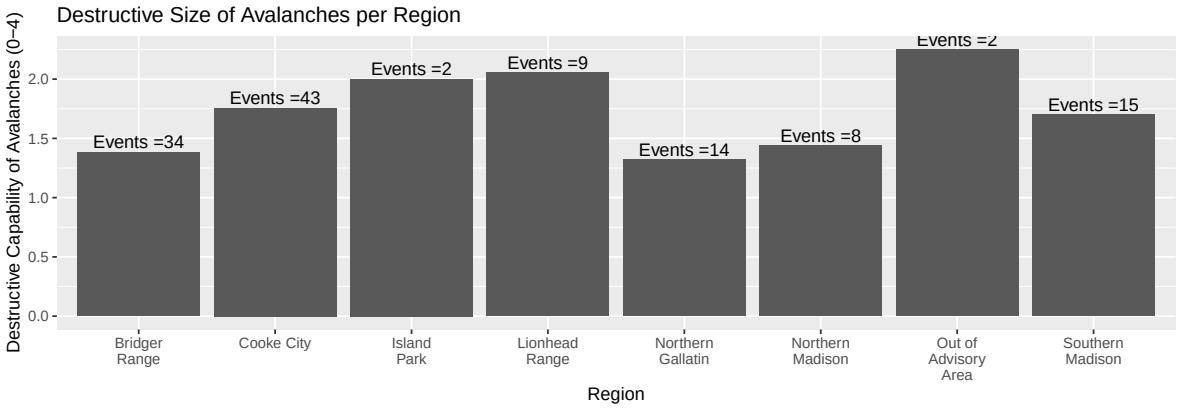
```
#plot avalanches by region

region_ammt <- avy_map |> count(Region) #count the ammount of avalanches per region

# plot the ammount of avalanches per region

region_plot <- ggplot(region_ammt, aes(
  x = Region,
  y = n
))+
  labs(
    title = "Avalanche Ammount Per Advisory Region",
    x = "Region",
    y = "Ammount of Avalanches"
      )+
  geom_col()+

  #stack x axis titles so they dont overlap

  scale_x_discrete(labels = function(x) str_wrap(x, width = 10))

region_plot
```



3B) Make a chart of average destructive capability (D) of avalanches in each region.

Avalanche size measured on the D scale describes the destructive capability of avalanches on a scale of 0 to 4. 0 indicates no destructive capability while size 4 can destroy 10 acres of Forrest and involves about 10,000 tons of snow. The following plot visualizes the average destructive capability (D) of avalanches in each region.

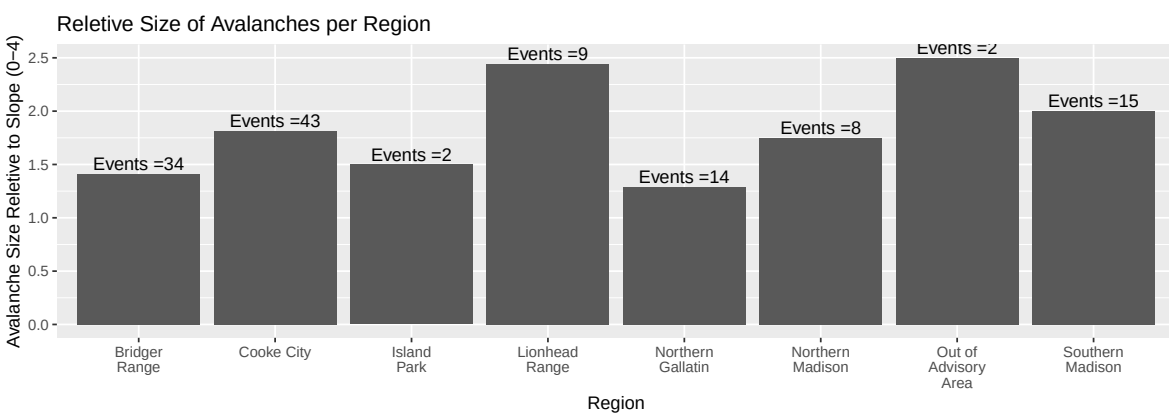
```
# compare aspect to D-size

D_avg <- avy_map %>% #take the everage destructive size (D.size) for each region
  group_by(Region) %>%
  summarise(mean_d = mean(D.size, na.rm = TRUE),
            n = n()
  )

#plot the average destructive size for avalanches in each region

D_plot <- ggplot(D_avg, aes(
  x = Region,
  y = mean_d
))+
  labs(
    title = "Destructive Size of Avalanches per Region",
    x = "Region",
    y = "Destructive Capability of Avalanches (0-4)"
      )+
  geom_col()+
  scale_x_discrete(labels = function(x) str_wrap(x, width = 10))+ #stack titles on the x
  geom_text(aes(label = paste0("Events =", n)),
            vjust = -0.3)

D_plot
```


Destructive Size of Avalanches per Region
Destructive Capability of Avalanches (0–4)
2.0
1.5
1.0
0.5
0.0
Events =34
Events =43
Events =2
Events =9
Events =14
Events =8
Events =2
Events =15
Bridger Range
Cooke City
Island Park
Lionhead Range
Northern Gallatin
Northern Madison
Out of Advisory Area
Southern Madison
Region

3C) Make a chart of average relative size (R) of avalanches for each region.

Similar to the D scale, the R scale measures the size of an avalanche. R stands for relative and measures the avalanche size relative to the maximum possible size the slope is capable of. R is measured on a scale of 1 to 4; 4 being the largest possible avalanche for a specific slope.

```
R_avg <- avy_map %>% #take the average destructive size (D.size) for each region
  group_by(Region) %>%
  summarise(mean_r = mean(R.size, na.rm = TRUE),
            n = n()
  )


#plot the average destructive size for avalanches in each region

R_plot <- ggplot(R_avg, aes(
  x = Region,
  y = mean_r
))+
  labs(
    title = "Reletive Size of Avalanches per Region",
    x = "Region",
    y = "Avalanche Size Reletive to Slope (0-4)"
      )+
  geom_col()+
  scale_x_discrete(labels = function(x) str_wrap(x, width = 10))+ #stack titles on the x
  geom_text(aes(label = paste0("Events =", n)),
            vjust = -0.3)


R_plot
```



Section 4: Make some tables

```
#remove geometry collumn

D_avg <- st_drop_geometry(D_avg)
R_avg <- st_drop_geometry(R_avg)
region_ammt <- st_drop_geometry(region_ammt)

# This simply calls a chart created earlier and uses gt() to make an aestetic chart.

D_avg %>%
  gt() %>%
  tab_header(title = "Average Avalanche Size (D)")
```

```
R_avg %>%
  gt() %>%
  tab_header(title = "Reletive Average Avalanche Size (R)")
```

```
region_ammt %>%
  gt() %>%
  tab_header(title = "Avalanches per region")
```

Average Avalanche Size (D)

Region	mean_d	n
Bridger Range	1.382353	34
Cooke City	1.755814	43
Island Park	2.000000	2
Lionhead Range	2.055556	9
Northern Gallatin	1.321429	14
Northern Madison	1.437500	8
Out of Advisory Area	2.250000	2
Southern Madison	1.700000	15

Reletive Average Avalanche Size (R)

Region	mean_r	n
Bridger Range	1.411765	34
Cooke City	1.813953	43
Island Park	1.500000	2
Lionhead Range	2.444444	9
Northern Gallatin	1.285714	14
Northern Madison	1.750000	8
Out of Advisory Area	2.500000	2
Southern Madison	2.000000	15

Avalanches per region

Region	n
Bridger Range	34
Cooke City	43
Island Park	2
Lionhead Range	9
Northern Gallatin	14
Northern Madison	8
Out of Advisory Area	2
Southern Madison	15

Analysis

There appears to be no correlation between D size and R size. For all of the regions, the D size and R size appear to be very similar. Considerations that might throw off the data would be a lack thereof. For example, notice how "Outside of Advisory Area" and "Island Park" only have two observations that appear to be quite large this may throw off data because those regions may not be as patrolled due to their distance. Furthermore, the large amount of observations in the Bridger and Cook City regions can likely be attested to their increased traffic and their proximity to Bozeman, where the GNFAC offices are.